

XITM6329 ASSIGNMENT - CENTRAL KAROO PROJECT

Scenario-Based Case Study: Central Karoo Interconnected Computer Network

AUTHORS

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Project Selection Process

According to Gido, Clement and Beker (2018), as the Project Managers for the Central Karoo District Community Network Implementation Project, our main aim is to choose a network solution that will provide reliable and affordable digital services to the rural communities. To select the best option, we followed four main steps from the project selection process

1. Set the criteria that will be used to compare the projects.
2. Identify the assumptions that will affect the project.
3. Compare the different network options using the selected criteria.
4. Select the network option that best meets the needs of the project.

Selection Criteria
To compare the two network options fairly, we selected six main criteria

1. Strategic Alignment
    - The project must support the goals of the Central Karoo District Municipality and national broadband plans.
2. Community Impact and Access
    - The network should provide access to hospitals, schools, municipal offices and residents in areas such as Beaufort West, Laingsburg and Prince Albert.
3. Cost Efficiency
    - The project should be affordable to build and maintain over time.
4. Technical Feasibility
    - The network must be suitable for the large distances, difficult terrain and weather conditions found in the Karoo.
5. Speed of Deployment
    - The network should be possible to build within the required project time frame.
6. Operational Resilience
    - The network should be able to continue working during load shedding, bad weather and other possible problems. It should also be easy to maintain over the long term.

Key Assumptions

1. Before comparing the options, we made the following assumptions
    - Municipal Co-operation: The municipality will provide the required site permissions, tower access and access to power where needed.
2. Spectrum Availability
    - The required ICASA approvals and wireless spectrum will be available within the planned project time.
3. Power Resilience
    - Solar panels and battery backup will be used at the main network sites so that the network can continue working during load shedding.
4. Community Support

- Local technicians will receive the necessary training to provide basic technical support within the district.

Evaluation Matrix

We compared two possible network solutions using the six selection criteria.
Scenario 1 a full fibre network would be installed to connect the main facilities and other locations across the district.
Scenario 2 a solar powered hybrid wireless network would be used for the main connections, while wireless towers and access points would be used to connect other locations and communities.
Each option was given a score from 1 to 5. A rating of 1 represents a poor result and a rating of 5 represents a very good result.

Selection Criteria	Weight	Project 1 Full Fibre rate	Weighted rate	Project 2 Hybrid Wireless rate	Weighted rate
Strategic Alignment	0.25	4	1.00	5	1.25
Community Impact and Access	0.20	3	0.60	5	1.00
Cost Efficiency and ROI	0.20	2	0.40	4	0.80
Technical Feasibility	0.15	3	0.45	4	0.60
Speed of Deployment	0.10	2	0.20	4	0.40
Operational Resilience	0.10	4	0.40	3	0.30
Total Score	1.00		3.05		4.35

Step 4: Project Selection Decision and Trade-Off Analysis

Project Decision Scenario 2 Hybrid Wireless Network
Based on the scores above, Scenario 2 received a total score of 4.35 out of 5 while Scenario 1 received 3.05 out of 5. Therefore, we would select Scenario 2 the Solar-Powered Hybrid Wireless Network for the Central Karoo District Community Network Implementation Project.

The decision is based on three main factors

Cost of Installing Fibre
Scenario 1 would require large amounts of fibre to be installed across the Central Karoo. This would involve digging and other construction work which could be expensive because of the area's rocky and difficult terrain. The high installation cost could also reduce the number of communities and households that can be connected.
Scenario 2 would use wireless links to connect locations over long distances. This would reduce the amount of physical construction needed and allow more of the project budget to be used to connect communities.

Faster Deployment
A full fibre network would take longer to install because of the construction work, approvals and other requirements involved. This could make it difficult to complete the project within the planned time frame.
The hybrid wireless option can be deployed more quickly because solar-powered towers and wireless access points can be installed at selected locations. This would allow important facilities such as hospitals municipal offices and community areas to receive connectivity sooner.
Flexibility and Community Coverage

A full fibre network mainly provides fixed connections. Extending the network to new or remote communities could require additional construction and installation work.
The hybrid wireless network provides more flexibility because additional wireless towers and access points can be added when needed. This makes it easier to expand the network as more households and communities require access.
Final Decision
Although the full fibre option may provide strong and reliable connections, the hybrid wireless network provides a better overall option for this project. It offers a good balance between cost, deployment speed, coverage and reliability. For the Central Karoo District, where the area is large and difficult to reach, Scenario 2 is therefore the most suitable option for meeting the project's goals.

Task 1: Overview Document

1. Selected Scenario-Based Description

The selected scenario addresses the severe digital infrastructure deficits of the Central Karoo District Municipality in the Western Cape, South Africa. Located approximately 417 km southeast of Cape Town, the region's local economy historically relied on the national railway junction. Following the systemic breakdown of the railway network, the population transitioned back to its foundational economic activity: agriculture.

The district's geographical expanse presents extreme terrain and weather challenges that severely hinder traditional telecommunications. While commercial mobile operators are progressively deploying 5G infrastructure, service availability remains highly volatile. Signal quality is frequently degraded by RF interference (noise) and sporadic connection drops, rendering commercial cellular networks unreliable for critical municipal and emergency services.To solve this, the Executive Mayor's Office is funding a public-sector project to design, deploy, and maintain the district's first interconnected community computer network.

2. Overview of the Purpose of the Selected Scenario

The primary purpose of this project is to build an independent, resilient, and secure high-speed network infrastructure tailored for a rural environment. Rather than depending on commercial cellular networks or single-point ISPs, this project establishes a hybrid intranet/internet framework owned by the district.

Transforming healthcare delivery for rural hospitals which cannot afford downtime when querying patient databases or transferring diagnostic images. If an emergency patient in Laingsburg needs transfer to Beaufort West, medical records can be synchronized over a secure, dedicated intranet pipe provided by Smart Bridges wireless links rather than waiting for unstable commercial ISP connections.

Securing government ERP systems and data trust that will Allow 200 government desktop users and external suppliers onto an unsegmented network creates severe cybersecurity risks. The initiative will introduce Network Access Control (NAC), Virtual Local Area Networks (VLANs), and Role-Based Access Control (RBAC). External suppliers will be restricted to isolated vendor subnets via secure VPNs, preventing them from probing or extracting municipal financial records stored on the ERP system.

Sustaining connectivity through renewable energy power instability in South Africa often causes standard cell towers and network nodes to shut down during extended blackouts. As the project leverages the existing NOC solar grid 70 deep-cycle batteries (200 Ah) coupled with 24 solar panels and power stabilizers ensuring that network routing, data transfer, and emergency service communications remain online 24/7, completely independent of the main electricity grid.

Bridging the Socioeconomic Digital Divide where Agriculture-based communities require reliable internet access to participate in modern economic and educational opportunities. By routing public Wi-Fi signals via the omnidirectional antenna systems at designated community hubs, residents across 3000 households gain access to municipal e-services, agricultural market prices, online education, and career opportunities.

3. Preliminary Overview of IT Solution

Beyond the issues outlined in the scenario, additional challenges may arise during implementation. These include potential network latency caused by long transmission distances between nodes, limited technical expertise among local staff to maintain advanced equipment, and regulatory constraints related to data protection and municipal IT governance. The proposed solution addresses these by incorporating automated monitoring tools, remote management capabilities, and compliance with South African POPIA data-privacy standards.

The current network suffers from limited coverage, inconsistent signal strength, and dependency on a single ISP, which restricts scalability and reliability. The proposed expansion to a mesh topology will eliminate single-point failures, improve bandwidth distribution, and ensure continuous connectivity even in adverse weather conditions. By integrating Smart Bridges multi-band access points, the network will achieve higher throughput and better interference management, supporting real-time hospital data access and administrative communication.

The decision to adopt a mesh or point-to-multipoint architecture is driven by the district's geographic dispersion and environmental constraints. Unlike the existing star topology, a mesh network provides dynamic rerouting capabilities, ensuring uninterrupted service when individual nodes fail. This design also simplifies future scalability, allowing new homes or institutions to be added without major infrastructure overhaul.

The preliminary IT solution involves leveraging and expanding the existing physical and network infrastructure to support a district-wide rollout. Given the environmental challenges (dust, heat) and the rural setting, the solution relies heavily on sustainable energy and wireless technologies. The solution will utilize the existing Network Operating Centre (NOC) as the central hub for data routing and power management. The expansion will require upgrading the wireless backbone infrastructure, scaling the Smart Bridges multi-band access points, and transitioning from a simple star topology to a more resilient mesh or expanded point-to-multipoint architecture to effectively cover the 3,000 residential homes.

4. Initial IT Description for the Selected Scenario

Although the current infrastructure is functional, it lacks redundant connectivity paths, cybersecurity measures, and disaster-recovery protocols. Introducing firewalls, intrusion-detection systems, and off-site data backups will strengthen resilience against cyber threats and hardware failure. Furthermore, implementing a cloud-based synchronization system will ensure that hospital and administrative data remain accessible even if local servers experience downtime.

The existing IT infrastructure is anchored by a purpose-built Network Operating Centre (NOC) designed with localized, natural cooling methods to combat dust and heat. The NOC is divided into four functional areas, a battery storage room, a server room, a working space, and an equipment storage room. While the NOC's current configuration provides a solid foundation, its limited cooling and power redundancy pose risks during peak load or extreme heat. The proposed upgrade will include intelligent power management systems and IoT-based environmental sensors to monitor temperature, battery health, and solar efficiency in real time. These additions will enhance operational resilience and reduce maintenance costs.

Power Infrastructure: The facility is heavily optimized for off-grid sustainability. It is powered by 24 roof-mounted solar panels (inclined at 20 degrees to minimize dust and maximize solar absorption, supported by a reinforced roof). The power is stored in a battery room containing seventy 200Ah deep-cycle batteries, managed by five inverters (including one pure sine wave inverter), two solar regulators, power stabilizers, and AC/DC disconnects.

Network and Hardware Infrastructure: The server room, designed without windows and facing south for optimal temperature control, houses a server rack and cooling fans. The physical network relies on a wireless backbone utilizing Smart Bridges with multi-band access points. The current network topology is a star configuration utilizing two main access points: one equipped with a 90-degree sectoral antenna for targeted directional coverage, and another with an omnidirectional antenna for broader, surrounding coverage.

The improved infrastructure will deliver a unified, high-availability network capable of supporting critical applications such as ERP, telemedicine, and secure data exchange. By integrating automated failover mechanisms and remote monitoring tools, the district will gain a self-sustaining IT environment that minimizes downtime and maximizes service delivery efficiency.

5. Business Process Alignment

The proposed Central Karoo Interconnected District Network will support the district's existing administrative, healthcare, and community processes while addressing the limitations of the current infrastructure. The existing network already supports approximately 200 government employees using Microsoft Office and the municipal ERP system, while the three district hospitals depend on reliable connectivity to access and share patient information across wards and administrative offices.

However, the current infrastructure creates several operational challenges. The hospitals are dependent on a local ISP, which can result in bottlenecks and unreliable access to patient information. The existing star topology also provides limited redundancy, meaning that a failure at a central network point could disrupt multiple users and services. Furthermore, expanding connectivity to approximately 3,000 homes could create congestion if the existing infrastructure is not upgraded.

The proposed solution will address these issues while maintaining the existing systems. The NOC, solar and battery infrastructure, ERP system, government computers, and hospital systems will continue to be used, but the network backbone will be upgraded to improve capacity and reliability. A mesh or expanded point-to-

multipoint architecture will provide alternative communication paths, reducing the impact of individual network failures.

Network segmentation through VLANs, NAC, RBAC, and secure VPNs will also ensure that users only access the systems required for their roles. Government employees can continue using the ERP system, healthcare staff can securely access authorised patient information, and external suppliers and community users can be isolated from sensitive municipal and healthcare systems.
Before implementation, network interruptions, ISP dependency, limited redundancy, and insufficient coverage can affect administrative and healthcare operations.

After implementation, the district will have a more reliable and scalable network that supports government administration, hospital services, and community connectivity. The existing infrastructure will therefore be maintained and improved rather than completely replaced, allowing the district to expand its digital services while reducing downtime and improving service delivery.

6. Potential Risks

1. Centralized Single Point of Failure (Availability Risk)

- Before: Currently, there are fragmentation issues with network reliability, and the communication services are frequently affected by loud noises and sporadic disconnections caused by the erratic 5G network. One breach of connection usually impacts only a localized area.
- After: By consolidating both routing and power management functions in the running Network Operating Centre (NOC), the project introduces a single point of failure. In case of an unfortunate situation like that of NOC facing a technical failure with the core routers, five inverters or the storage device, the entire block would go dead. It would lead to the breakdown of 3,000 residential houses, administrative offices, and services provided by the three hospitals located in the block.

2. Expanded Attack Surface (Information Security Risk)

- Before: The three district hospitals currently rely on a local Internet Service Provider (ISP) to access patient data and maintain communications. The data flows through a relatively restricted, albeit bottlenecked, pipeline.
- After: Integrating 3,000 residential homes together with an administrative building containing 200 governmental employees, and hospital databases in unified infrastructure results in the considerable expansion of the attack surface of the network. Lack of a robust logical segmentation, including VLANs, and access control makes it possible for an attacker to access a residential node and switch to the administrative network thereby assuming control of ERP or obtaining patients' records.

3. Physical Security and Asset Management Risk

- Before: High-value IT assets are primarily secured and concentrated in controlled environments - specifically, the administrative building's 28 offices and the hospitals' 40 wards and executive offices.
- After: The purchasing and widespread implementation of high-capacity wireless access points, routers, and Customer Premises Equipment (CPE) to 3,000 rural houses introduces serious physical security breaches. The spread of hardware in the open air or in residential areas is very likely to face theft, vandalism or

accidental destruction. Fast replacement of damaged items would quickly use up the 10% contingency budget set aside for unexpected malfunctions.

4. Operational Skill Deficit and Maintenance Burden

- Before the municipality has outsourced complex routing and connectivity issues to the local internet service provider, which runs the entire infrastructure for the hospitals.

- After Moving to a district-wide network controlled internally means that local IT staff carry the entire operational burden. There is an allocation of 15% of the budget for specialized training to fulfill the perimeter and the network operation center. However, managing complex routing can be labor-intensive because it often entails managing mesh or point-to-multipoint networks with thousands of endpoints. If the IT staff fails to keep pace with the technology, the network will be down for a long period of time.

5. Quality of Service (QoS) and Scope Creep

- Before: The government workforce needs to barely concern itself with the requirement of dealing only with municipal projects through Microsoft Office and an Enterprise Resource Planning system.

- After: As three thousand households are connected to the network, considerable unpredictable residential internet traffic (for example, media streaming) will be generated. If no strict Quality of Service is introduced to give priority to accessing hospital data and administrative functions of the ERP system, the bandwidth will be blocked to an extent that critical public services would be working inefficiently.

7 Cost Estimations

Based on initial budgeting caps provided by the mayor's office, the project has an estimated budget ceiling of approximately R2,000,000. The cost estimations are divided into the following categories:

a. Machinery and Infrastructure (~R900 000): Finding additional servers, wire-capacity access points, routers, and Customer Premises Equipment for the 3,000 homes. Upgrades to the battery selection may also be needed.
b. People and Labor (~R600 000): Costs related to network engineers, installation technicians for residential placements, project managers, and initial consulting fees.
c. Training and Handover (~300 000): Specialized training for local IT staff to manage the expanded network and NOC, and basic digital literacy support for community integration.
d. Contingency (~200 000): Reserved for unexpected environmental mitigations or hardware failures during rollout.

Additional costs consideration:

a. Network Scalability: Existing network was not made for the 3000 new households therefore adding capacity will be necessary to avoid things like congestion and the with the additional capacity the network will keep performing well even if the number of users increase.
b. Environmental Safety: The wireless equipment outside might be exposed to things like dust, heat, wind and other conditions. Using the proper outdoor equipment can increase the wireless equipment’s duration.
c. Community Support: Because this project enhances the wireless connectivity to 3000 homes or more the resources may also be needed for simple user support and teaching the locals new network services.

1. Project Summary and Title

Central Karoo District Community Network Implementation Project

This project charter outlines the framework for designing, implementing, and managing a district-wide interconnected computer network in the Central Karoo. Sponsored by the Mayor's Office, the project will upgrade the existing Network Operating Centre (NOC) and wireless backbone to provide stable, high-speed connectivity to three district hospitals, one main administrative building, and ultimately, up to 3,000 residential homes, overcoming the region's historical connectivity deficits.

2. Project Purpose, Goals, and Objectives

The Central Karoo district is suffering from unreliable communications infrastructure as they rely on commercial ISPs. The region lacks fixed telephony, while commercial 5G rollouts exhibit severe instability, frequent signal dropouts, and heavy noise. Local institutions, including three district hospitals managing 40 wards and 10 executive offices each and a municipal administrative building housing 200 employees across 28 offices still rely on commercial ISPs. Which creates unwanted network bottlenecks and delays with access to critical patient records. With security and trust, due to third-party suppliers having access or accessing government software systems.

This project delivers a municipal-owned, independent wireless backbone network powered by sustainable solar infrastructure. Over a 12-month timeline, the initiative will provide reliable intranet and internet access to public sector nodes and approximately 3,000 community households, improving service delivery and regional connectivity.

Project goals

- Transition public sector operations away from third-party commercial Internet Service Providers ISPs by deploying a municipality owned self-sustained wireless backbone infrastructure. Which will in return establish an independent municipal network.
- Enhance Public Service Delivery by introducing and enabling reliable access to real-time administrative data and critical electronic patient records across health facilities and government entities throughout the region to ensure healthcare access for everyone in the community.
- Bridge the digital divide gap in rural Communities to deliver accessible intranet and internet connectivity to underserved rural residents across the Central Karoo district.
- Ensure operational sustainability and data integrity by building an eco-friendly power-backed network footprint while securing government software systems against untrusted third-party access.

Project objectives

1. Infrastructure & Connectivity Deployment

- Construct and operationalize a dedicated point-to-multipoint outdoor wireless backbone within a strict 12-month timeline.
- Provide stable intranet and internet access to 3,000 community households.
- Link 3 district hospitals encompassing 40 wards and 10 executive offices each and 1 municipal administrative building housing 200 employees across 28 offices to the unified backbone.

2. Network Performance & Reliability

- Eliminate frequent signal dropouts, connectivity bottlenecks, and heavy network noise caused by existing commercial cellular/ISP links.
- Reduce latency and packet loss across health network channels to accelerate inter-ward and inter-hospital patient record transfers.
- Power the network using sustainable solar infrastructure to maintain continuous uptime in remote locations.

3. Security & Administrative Control

- Safeguard internal government applications such as ERP software by restricting and monitoring third-party supplier access.
- Establish network segmentation and strict access controls to resolve trust concerns surrounding external vendor access to sensitive public data.

3. Project Scope and Milestones

In Scope: Upgrading existing NOC infrastructure; deploying new multi-band access points; configuring the network to support ERP and MS Office traffic for 200 govt users; installing receivers at 3,000 homes; providing IT training for NOC operators.

The defined scope ensures that the project remains focused on infrastructure and connectivity improvements rather than peripheral upgrades. By excluding concentrated resources on the network backbone, which is the most critical enabler of service delivery. Each milestone is strategically sequenced to minimize disruption to hospital and administrative operations while ensuring progressive community rollout. Department between phases are managed through continuous stakeholder engagement and risk reviews at the end of each milestone.

Out of Scope: Upgrading end-user devices (desktops/laptops) within the hospitals or administrative building; providing power backup solutions for the residential homes; developing custom software for the municipality.

Phase	Milestone	Estimated Completion
1	Project Kick-off & Stakeholder Agreement	Month 1
2	NOC Assessment & Infrastructure Procurement	Month 2
3	Core Network Upgrade & Admin/Hospital Integration	Month 5
4	Residential Rollout (Phase A - 1000 homes)	Month 8
5	Residential Rollout (Phase B - 2000 homes)	Month 12
6	Final Testing, Training & Handover	Month 13

The milestone structure reflects a phased implementation approach that balances technical complexity with community impact. Early phase focus on assessment and procurement to establish a stable foundation, while late phases emphasize scalability and user adoption. This progression supports effective monitoring allowing lessons learned in Phase 1 to inform Phase 2, thereby reducing deployment risks and optimizing resource utilization.

4. Key Deliverables, Timing, and Resources

The successful execution of this project relies on the following key deliverables (KD), bound by specific timelines and resource allocations:

KD1: Network Architecture Design Document. (Timing: Month 1). Resources: Senior Network Architect, IT Management Group.
KD2: Upgraded NOC Hardware Installation. (Timing: Month 3). Resources: Hardware technicians, existing solar/battery power grid.
KD3: Hospital Data Migration Plan. (Timing: Month 4). Resources: Database administrators, ISP liaison, Hospital IT leads.
KD4: Community Access Nodes Deployment. (Timing: Months 6-12). Resources: Field installation teams, Mayor's community outreach office.

Each deliverable directly contributes to the projects overarching goal of establishing a resilient, self-sustaining network. The Network Architecture Design Document provides the blueprint for all subsequent activities, ensuring technical alignment and compliance with municipal IT standards. The Upgraded NOC Hardware Installation strengthens the districts core infrastructure, enabling higher healthcare information during transition, maintaining operational continuity. Finally, the Community Access Node Deployment Extends connectivity to residents, promoting digital inclusion and socio-economic development.

Resource allocation emphasizes collaboration between technical experts and local stakeholders. By leveraging existing solar infrastructure and community outreach programs, the leveraging existing solar infrastructure and community outreach programs, the project maximizes cost efficiency and sustainability. Regular progress audits will ensure that deliverables remain on schedule and within the R2 million budget cap

5. Budget and Cost-Benefit Analysis

The Central Karoo Interconnected District Network has an estimated budget ceiling of R2,000,000. The budget is allocated to Infrastructure and Machinery (R900,000), People and Labour (R600,000), Training (R300,000), and a Contingency Reserve (R200,000). This allocation ensures that sufficient resources are available for network deployment, technical expertise, staff training, and unexpected costs.

From a cost-benefit perspective, the project provides both immediate and long-term value. Before implementation, the hospitals and government offices depend on the existing ISP and infrastructure, which can result in recurring connectivity costs, bottlenecks, and potential downtime. Limited infrastructure also restricts the district's ability to expand connectivity to the wider community.

After implementation, the upgraded network will reduce dependence on the existing ISP for critical government and hospital connectivity while improving reliability and scalability. The existing NOC and renewable-energy infrastructure will also be retained, allowing the municipality to build on its previous investment rather than replacing the entire system.

The investment will additionally provide connectivity to up to 3,000 residential homes, supporting access to municipal e-services, education, employment opportunities, and agricultural information. Although the project requires significant upfront expenditure, these benefits can improve service delivery, reduce operational disruption, and support local economic development over the long term. Therefore, the R2 million investment is justified by the expected improvements in connectivity, operational efficiency, community access, and network resilience.

6. Stakeholders

Project Sponsor: The Mayor's Office (Has the mandate to issue funding and establish the project strategy). They have control over the R2,000,000 funding limit and the strategic vision of the project. They also require the provision of executive summaries that include details about budget utilization and accomplishment of milestones.
Group from the IT Management Department (consisting of students and consultants) - This provides planning, advice, and approves the task.
Lecturer / Academic Assessor: Confirms the project's compliance with academic requirements. It is absolutely necessary to comply with the course rubric throughout the project.
Hospital Administrators: Main users of the system whose requirements include secure data preserving zero downtime during the migration since the absence of access to data might influence the provision of medical treatment.

External Suppliers: Hardware suppliers, who are going to supply additional servers, Smart Bridges hardware, Customer Premises Equipment (CPE). All procurement steps should be planned in order to avoid bottlenecks in the supply chain during Phase 2.
The current local ISP (must cooperate during the transition phase) should take part in the transition process. This is obligatory during the transition phase, as any conflicts in relationships may lead to the disconnection of the hospital's communications system
Community Representatives: Representing the interests of the 3,000 homes being connected. They require clear communication and updates regarding the phased implementation dates to manage public expectations regarding service delivery.
Agreement Process: The charter will be distributed to all primary stakeholders. There will be a formal review meeting to discuss concerns. Once there is consensus, the Mayor's office, the head of IT Management, and representatives of the hospital, will endorse the document and approve the commencement of the project..

7. Training and Handover

To ensure the sustainability of the network post-implementation, a comprehensive training program is included. This involves:

A. Technical Training: Equipping local IT staff with the skills to monitor the network using the NOC, troubleshoot wireless link failures, and maintain the solar/battery power infrastructure.

B. End-User Training: Briefing the 200 government employees on the new network protocols and security measures when accessing the ERP and MS Office systems.

C. Train-to Trainer and Documentation: this is the process of selecting, upskilling along with checking whether trainers are certified to assess the hiring process of hiring new staff using the run books, manuals, handover packs and version control to make sure that operations run efficiently minimizing human error.

D. Handover Mechanics and Governance: The process of accepting formal criteria by defining calculated items, with the checklist to show verified authority, escalation matrix for help clear the lines when escalation is needed, legal and compliance briefing to provide a summary of ICT regulations that the governance can regulate, and inventory transfer do physical handovers of spare parts and other required equipment with the correct details like warranties.

E. Success Metrics and acceptance criteria: This involves the operations readiness so that the Network Operations Centre employees can solve common problems alone and within the target times, along with the competence of users because a large percentage of trained employees can finish practical ERP tasks, and the documents completeness will involve the SOPs, inventories and pave new ways that can be documented.

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